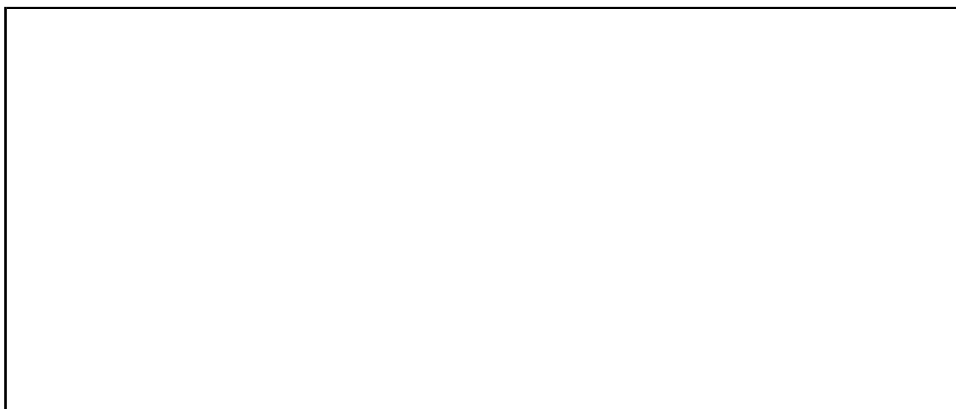


CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Fourth Semester B.Sc.(Physics) Examination April-May 2018
PD 251 Mathematical Physics-III

Date: 30/04/2018, Monday **Time:** 01:30 p.m. to 02:00 p.m. **Maximum Marks:** 20

Multiple choice questions



Important Instructions:

- Provide all necessary information as required in the answer sheet.
 - Attempt all questions.
 - Answer the questions in the question paper itself.
 - Use of cell phones is strictly prohibited
 - Use of non-programmable calculators may be allowed if needed.
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Q.1. Choose the correct answer for the following questions. [20]

1. If $L[f(t)]$ is the Laplace transform of $f(t)$, then $L[te^{2t}] =$ _____.

- a. $\frac{2}{s^2}$ b. $\frac{1}{(s-2)^2}$ c. $-\frac{1}{(s-2)^2}$ d. $\frac{1}{(s-2)^3}$

2. If $L^{-1}[\bar{f}(s)]$ is the inverse Laplace transform of $\bar{f}(s)$, then $L^{-1}[e^{-5s}] =$ _____.

- a. $\delta(t-5)$ b. $u(t-5)$ c. e^{5t} d. $\frac{1}{t-5}$,

where $u(t-a)$ is the unit step function and $\delta(t-a)$ is the Dirac delta function.

3. $L[t^3] =$ _____.

- a. $\frac{24}{s^3}$ b. $\frac{24}{s^4}$ c. $\frac{6}{s^4}$ d. $\frac{6}{s^3}$.

4. $L^{-1}[\frac{1}{s^2-7}] =$ _____.

a. $\cosh \sqrt{7}t$

c. $\frac{\cosh \sqrt{7}t}{\sqrt{7}}$

b. $\sinh \sqrt{7}t$

d. $\frac{\sinh \sqrt{7}t}{\sqrt{7}}$

5. If $F(s)$ is the Fourier transform of function $f(x)$, then _____.

a. $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s)e^{-isx} ds$

c. $f(x) = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} F(s)e^{-isx} ds$

b. $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) \cos sx ds$

d. $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) \sin sx ds$

6. If $F[f(x)] = F(s)$ is the Fourier transform of $f(x)$, then $F[f(x-a)] =$ _____.

a. $e^{-ias}F(s)$

c. $e^{as}F(s)$

b. $e^{ias}F(s)$

d. $e^{-as}F(s)$.

7. If $F[f(x)]$ is the Fourier transform of $f(x)$ and $f(x) * g(x)$ denote the convolution product of $f(x)$ and $g(x)$, then $F[f(x) * g(x)] =$ _____.

a. $F[f(x)] \cdot F[g(x)]$

c. $F[f(x)] - F[g(x)]$

b. $F[f(x)] + F[g(x)]$

d. $F[f(x)] * F[g(x)]$.

8. If the Fourier sine transforms of $f(x)$ is $F_s(s)$, then $\frac{2}{\pi} \int_0^{\infty} |F_s(s)|^2 ds =$ _____.

a. $\frac{2}{\pi} \int_0^{\infty} |f(x)|^2 dx$

c. $\int_{-\infty}^{\infty} |f(x)|^2 dx$

b. $\frac{2}{\pi} \int_{-\infty}^{\infty} |f(x)|^2 dx$

d. $\int_0^{\infty} |f(x)|^2 dx$

9. If $i = \sqrt{-1}$, then $i^{49} =$ _____.

a. i

b. $-i$

c. 1

d. -1 .

10. What is the principal argument of the complex number $-\sqrt{3} - i$?

a. $\frac{\pi}{4}$

b. $\frac{3\pi}{4}$

c. $\frac{\pi}{6}$

d. $\frac{3\pi}{2}$.

11. The function $f(z) = |z|^2$ is _____.

a. nowhere analytic

c. analytic only at $z = 1$

b. analytic only at origin

d. an entire function

12. Which of the following function is harmonic?

- a. $x^2 + y^2$ b. $\cos xy$ c. $e^x \cos y$ d. $\sin xy$.

13. If $z = r(\cos \theta + i \sin \theta)$, then $|z|^3 =$ _____.

- a. $(\cos \theta + i \sin \theta)^3$ c. r^3
b. $r^3(\cos \theta + i \sin \theta)^3$ d. $\frac{r^3}{2}$

14. If $\int_{|z|=1} \frac{dz}{z-7} =$ _____.

- a. $14\pi i$ b. $2\pi i$ c. $7\pi i$ d. 0

15. According to Liouville's theorem, bounded entire function is _____.

- a. non constant c. not analytic
b. constant d. discontinuous.

16. The power series $\sum_{n=0}^{\infty} z^n$ converges for _____.

- a. all $z \in \mathbb{C}$ c. $|z| = 1$
b. $|z| < 1$ d. none of these.

17. The function $z \cos\left(\frac{1}{z}\right)$ has _____ at $z = 0$.

- a. removable singularity c. simple pole
b. essential singularity d. double pole.

18. What is the residue of function $ze^{\frac{1}{z}}$ at origin?

- a. 1 b. 2 c. $\frac{1}{2}$ d. 0

19. What is the residue of $\frac{1-2z}{z(z-1)(z-2)}$ at $z = 2$?

- a. $\frac{1}{2}$ b. 1 c. $\frac{3}{2}$ d. $-\frac{3}{2}$

20. $\int_{|z|=3} \frac{e^z}{z+1} dz =$ _____.

- a. 0 b. πe c. $\frac{2\pi i}{e}$ d. $2\pi i e$.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Fourth Semester B.Sc.(Physics) Examination April-May 2018
PD 251 Mathematical Physics-III

Date: 30/04/2018, Monday **Time:** 02:00 p.m. to 4:30 p.m. **Maximum Marks:** 50

Instructions:

1. Section I and II must be attempted in separate answer sheets.
 2. Make suitable assumptions and draw neat figures wherever required.
 3. Use of non-programmable calculator is allowed.
 4. Show necessary calculations.
 5. Figures to right indicate marks.
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SECTION-I

Q-2 Answer the following questions as directed. [20]

(a) Define unit step function and find its Laplace transform. [02]

(b) Find the Laplace transform of the periodic function [03]

$$f(t) = \begin{cases} \cos \omega t, & \text{for } 0 < t < \frac{\pi}{\omega}, \\ 0, & \text{for } \frac{\pi}{\omega} < t < \frac{2\pi}{\omega}, \end{cases} \quad \text{with period } T = \frac{2\pi}{\omega}.$$

OR

(b) Find the Laplace transform of $\frac{\cos 3t - \cos 2t}{t}$. [03]

(c) Find an analytic function whose real part is $x^2 - y^2$. [03]

OR

(c) Show that a function $\log(x^2 + y^2)$ is harmonic. [03]

(d) Prove that $\int_C \frac{dz}{z-5} = 2\pi i$, where C is the circle $|z-5| = 2$. [02]

OR

(d) Find the square root of the complex number $5 + 12i$. [02]

(e) Express $\frac{(6+i)(2-i)}{(4+3i)(1-2i)}$ in the form of $a + ib$. [02]

(f) Define inverse Fourier cosine transform. Hence solve the integral equation [03]

$$\int_0^{\infty} f(x) \cos sx \, ds = \begin{cases} 1-s, & \text{for } 0 \leq s \leq 1, \\ 0, & \text{for } s > 1. \end{cases}$$

OR

(f) Find the function $f(x)$ whose Fourier sine transform is $\frac{e^{-5s}}{s}$. [03]

(g) Define Dirac delta function. Express Dirac delta function as a Fourier integral. [03]

OR

(g) Express the function $f(x) = \begin{cases} 1, & \text{for } |x| \leq 1, \\ 0, & \text{for } |x| > 1, \end{cases}$ as a Fourier integral. [03]

(h) Find the Fourier sine transform of $f(x) = \begin{cases} 4x, & \text{for } 0 < x < 1, \\ 4 - x, & \text{for } 1 < x < 4, \\ 0, & \text{for } x > 4. \end{cases}$ [02]

SECTION II

Q-3 Answer the following questions as directed. [30]

(a) Expand $f(z) = \frac{1}{(z+2)(z+4)}$ in Laurent series valid for (a) $|z| < 2$, [05]
(b) $2 < |z| < 4$ and (c) $|z| > 4$.

(b) Using Cauchy's residue theorem, evaluate the integral $\int_0^{2\pi} \frac{d\theta}{5 - 3 \cos \theta}$. [05]

OR

(b) Evaluate the integral $\int_C \frac{z^2}{(z-1)^2(z+2)} dz$, where C is the circle $|z| = 3$. [05]

(c) Find the roots of the equation $z^6 + i = 0$. [05]

OR

(c) Prove that $\cos z$ is an entire function. [05]

(d) Solve the one dimensional heat equation $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$, for $0 \leq x < \infty, t > 0$ [05]
subject to conditions (i) $u(x, 0) = 0$, for $x \geq 0$ (ii) $\frac{\partial u}{\partial x}(0, t) = -a$,
(iii) $u(x, t)$ is bounded, where k and a are constants.

OR

(d) Define Fourier transform of function $f(x)$. Find the Fourier transform of the [05]
Gaussian distribution function $f(x) = ke^{-lx^2}$, where k and l are constants.

- (e) Using Laplace transform, solve the simultaneous equations: [05]

$$\frac{dx}{dt} + y = \sin t, \quad \frac{dy}{dt} + x = \cos t,$$

given that $x(0) = 2$ and $y(0) = 0$.

OR

- (e) Using the convolution theorem, find the inverse Laplace transform of [05]

$$\frac{s^2}{(s^2 + 25)(s^2 + 9)}.$$

- (f) A particle moves in a line so that its displacement y from a fixed point O at any time t , is given by $\frac{d^2y}{dt^2} - 7\frac{dy}{dt} + 10y = 8e^{-t}$. Using Laplace transform, find its displacement at any time t if initially particle is at rest at $y = 0$. [05]

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